

Packet Tracer – Inter-VLAN Routing Using Router-on-a-Stick

Objectives

Part 1: Build a Multi-VLAN Network Topology

Part 2: Create and Assign VLANs

Part 3: Configure a Trunk Link

Part 4: Configure Router-on-a-Stick

Part 5: Verify Inter-VLAN Connectivity

Background / Scenario

In a switched network, devices in different VLANs are separated into different broadcast domains and cannot communicate directly. To allow communication between VLANs, a Layer 3 device must perform routing.

In this lab, you will configure **Router-on-a-Stick**, a common method of inter-VLAN routing where a single router interface uses multiple subinterfaces to route traffic between VLANs.

This lab focuses on:

- Creating VLANs
- Assigning switch ports to VLANs
- Configuring 802.1Q trunking
- Configuring router subinterfaces
- Verifying inter-VLAN communication

You are building a core CCNA skill used in enterprise networks to separate departments while still allowing controlled communication between them.

Instructions

Part 1: Build the Network Topology

Step 1: Add the Devices

a. Open **Cisco Packet Tracer**.

b. Add the following devices:

- 1 Router (1941 or 2911)
- 1 Switch (2960)
- 2 PCs

c. Rename the devices:

- Router0 → **R1**
 - Switch0 → **S1**
 - PC0 → **Sales-PC**
 - PC1 → **HR-PC**
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Step 2: Connect the Devices

Use **Copper Straight-Through** cables.

Connect:

Device	Interface	Device	Interface
Sales-PC	FastEthernet0	S1	FastEthernet0/1
HR-PC	FastEthernet0	S1	FastEthernet0/2
R1	GigabitEthernet0/0	S1	GigabitEthernet0/1

Wait until the links turn green.

Your physical topology is now complete.

Part 2: Configure VLANs on the Switch

Step 1: Access the Switch CLI

Open the CLI of **S1** and enter:

enable

configure terminal

Step 2: Create VLAN 10

Type:

```
vlan 10
name SALES
exit
```

Step 3: Create VLAN 20

Type:

```
vlan 20
name HR
exit
```

Step 4: Assign Ports to VLANs

Configure FastEthernet0/1 for VLAN 10:

```
interface fastethernet 0/1
switchport mode access
switchport access vlan 10
exit
```

Configure FastEthernet0/2 for VLAN 20:

```
interface fastethernet 0/2
switchport mode access
switchport access vlan 20
exit
```

Step 5: Verify VLAN Configuration

Type:

```
show vlan brief
```

Expected output:

VLAN Name	Status
10 SALES	active
20 HR	active

You should see Fa0/1 in VLAN 10 and Fa0/2 in VLAN 20.

Part 3: Configure the Trunk Link

The router must receive traffic from multiple VLANs over a single interface.

Step 1: Configure the Trunk Port

On S1, enter:

```
interface gigabitethernet 0/1
switchport mode trunk
exit
```

Step 2: Verify the Trunk

Type:

```
show running-config
```

Expected result:

```
interface gigabitethernet 0/1
switchport mode trunk
```

The trunk link is now ready.

Part 4: Configure Router-on-a-Stick

Step 1: Access the Router CLI

Open **R1** and enter:

```
enable
configure terminal
```

Step 2: Enable the Physical Interface

Type:

```
interface gigabitethernet 0/0  
no shutdown  
exit
```

Step 3: Configure Subinterface for VLAN 10

Type:

```
interface gigabitethernet 0/0.10  
encapsulation dot1Q 10  
ip address 192.168.10.1 255.255.255.0  
exit
```

Step 4: Configure Subinterface for VLAN 20

Type:

```
interface gigabitethernet 0/0.20  
encapsulation dot1Q 20  
ip address 192.168.20.1 255.255.255.0  
exit
```

Why Use Subinterfaces?

Each subinterface acts as the default gateway for a specific VLAN.

VLAN	Gateway
-------------	----------------

VLAN 10	192.168.10.1
---------	--------------

VLAN 20	192.168.20.1
---------	--------------

This allows the router to route traffic between VLANs.

Part 5: Configure the PCs

Step 1: Configure Sales-PC

Go to:

Desktop → IP Configuration

Enter:

- IP Address: 192.168.10.10
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.10.1
-

Step 2: Configure HR-PC

Go to:

Desktop → IP Configuration

Enter:

- IP Address: 192.168.20.10
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.20.1
-

Part 6: Verify Connectivity

Step 1: Test the Default Gateway

From Sales-PC:

ping 192.168.10.1

You should receive successful replies.

From HR-PC:

ping 192.168.20.1

You should also receive successful replies.

Step 2: Test Inter-VLAN Communication

From Sales-PC:

ping 192.168.20.10

Expected Result:

Reply from 192.168.20.10

You should receive 4 successful replies.

Step 3: Test from HR-PC

Type:

ping 192.168.10.10

You should again receive successful replies.

The router is now routing traffic between VLANs.

Understanding Router-on-a-Stick

Traffic flow:

Sales-PC
(VLAN 10)

|

|

S1

|

Trunk

|

R1 Gi0/0

| |

.10 .20

| |

VLAN10 VLAN20

|

|

S1

|

HR-PC

(VLAN 20)

The switch tags VLAN traffic using **802.1Q**, and the router routes packets between the VLANs.

Troubleshooting Tips

If the ping fails:

✓ Verify VLAN assignments using:

show vlan brief

✓ Verify trunking using:

show interfaces trunk

✓ Ensure the router interface is not administratively down.

✓ Verify subinterface IP addresses.

✓ Check default gateways on both PCs.

✓ Confirm VLAN IDs in the router match VLAN IDs on the switch.

✓ Wait a few seconds for interfaces to initialize.

What You Learned

- How VLANs create separate broadcast domains.
 - How to create and assign VLANs.
 - How to configure a trunk port.
 - How 802.1Q tagging works.
 - How Router-on-a-Stick enables inter-VLAN routing.
 - How to configure router subinterfaces.
 - How to verify communication between VLANs.
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Challenge Extensions (Optional)

Try modifying the lab:

1. Add **VLAN 30 (IT Department)** and connect another PC.
2. Configure an additional subinterface:

```
interface gigabitethernet 0/0.30
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0
```

3. Restrict communication between VLANs using an ACL.
4. Use Simulation Mode to observe VLAN tags and routing behavior.
5. Replace Router-on-a-Stick with a Layer 3 Switch and compare the configuration.